Question Paper

Exam Date & Time: 21-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (CSE STREAM) EXAMINATIONS - NOVEMBER 2024
SUBJECT: ECE 1072- FUNDAMENTALS OF ELECTRONICS

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A)	Obtain expressions for efficiency and ripple factors of full wave rectifier with ideal diodes. A sinusoidal voltage of 50 Sin(2π50t) V is applied to a full wave rectifier. If the load resistance is 1000Ω, determine the average and RMS value of load currents.	(4)
1B)	A Silicon diode carries a constant current of 1mA, when the operating temperature is 20°C for a diode voltage of 700mV. If the temperature rises to 40°C, what is the new value of diode voltage?	(3)
1C)	Consider a Zener voltage regulator with $V_Z = 10V$, $R_S = 1K\Omega$, $R_L = 2K\Omega$ and input voltage varying from 22V to 40V. Determine the maximum and minimum values of Zener current. Also, determine the maximum power dissipated by the Zener diode.	(3)
2A)	With a neat circuit diagram, explain the working principle of self-bias circuit using MOSFET. Consider a self-bias circuit using MOSFET with $R_1=4K\Omega$, $R_2=10K\Omega$, $R_S=1K\Omega$ and $V_{DD} = 1.8V$. Determine the bias current by assuming $V_{TH} = 0.5V$, $\mu_n C_{ox} = 100\mu A/V^2$ and W/L=5/0.18. Also, determine the maximum allowable value of drain resistance (R_D) that can maintain MOSFET in saturation region.	(4)
2B)	From the fundamentals, obtain an expression for the drain current of the MOSFET in terms of V_{GS}, V_{DS} and the dimension of the device.	(3)
2C)	Consider an OPAMP comparator with inverting input connected to ground (i.e $V_{inv}=0V$) and noninverting input supplied with a signal of V(t)=3sin(wt). Assuming that the OPAMP is connected to ±15V power supply, sketch the output waveform. Also, determine the output voltage if the inverting input voltage is changed to -5V (i.e $V_{inv} = -5V$).	(3)
3A)	With the help of a neat circuit diagram and relevant mathematical expressions, explain how an OPAMP can be used as a subtractor. Design an OPAMP difference amplifier circuit to obtain $V_{out}=2\,(V_2-V_1)$, where V_2 and V_1 are the inputs.	(4)
3B)	Subtract $(67)_8$ from $(67)_{10}$ using 2's Complement method	(3)
3C)	Encode the data 1001 using i) Gray Code. ii) (7, 4) even parity hamming code.	(3)
4A)	Reduce the following Boolean expression using K-map and realize using AOI logic. f = AB + AB'C + A'BC' + BC'	(4)
4B)	The logic system has 3 inputs and one output. The output is 1 if only one input is made high. Write the truth table. Write the expression for the output in SOP form.	(3)
4C)	Design a 3 - bit ripple up counter using negative edge triggered JK Flip Flops.	(3)
5A)	Design a SISO shift register using D flip-flops to store 4-bit data 0101. After how many cycles, MSB of data will be available at the output.	(4)
5B)	With a neat logic diagram using only NAND gates and truth table, explain the working of JK flip-flop.	(3)
5C)	Realize a 4-to-1 multiplexer using basic gates.	(3)

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